

Syllabus

Spring Semester 2026

Course Title	General Chemistry Laboratory 1	Course Code		(E)CB3102		Credit	1	Classroom	Tamgu-gwan 2503
		Type		Elective (AP)		Hours	2		
Teacher	Name	Class No.	Office	Office Hour*	Contact		E-mail		
	Yeosan Lee (이여산)	1, 2, EC	2505	Tuesday 1-4 periods	+82-51-606-2223		yeosanksa@ksa.hs.kr		

* Office Hour: Time for students to ask questions and discuss lecture or lab topics with teachers for deeper understanding.

1. Course Description

Chemistry is central to our understanding of the natural sciences. It examines the composition, structure, and properties of substances, as well as the transformations they undergo. General Chemistry Lab 1 is designed to provide students with a practical understanding of fundamental chemical principles and their quantitative application in the laboratory. The course covers topics ranging from basic laboratory techniques to synthetic reactions of organic and inorganic compounds introduced in the General Chemistry I lecture course. It also includes a project using a quantum chemistry calculation program.

2. Core Target Competencies

Subject Core Competencies						
Subjective Comprehension	Scientific Thinking	Problem Solving Ability	Communication Skills	Ability to Explore	Teamwork	Creativity
Through hands-on experiments that apply the knowledge and theories introduced in General Chemistry I, students deepen their understanding of course concepts while cultivating scientific thinking, inquiry skills, creativity, and collaboration. Team-based projects further strengthen scientific reasoning, communication, and teamwork.						

3. Textbooks and References

3.1. Textbook

(1) Laboratory Manual for Principles of General Chemistry, 10th Edition - by J. A. Beran (John Wiley & Sons, Inc.)

3.2. References

(1) Principles of Modern Chemistry, 8th Edition - by Oxtoby, Gillis, and Butler (Cengage Learning)

4. Class Method

4.1 Class Management

- (1) The course is based on student-centered learning activities.
- (2) Each student writes and submits a pre-lab report after studying about the topic to be experimented with in advance.
- (3) Two or three students form a team to conduct an experiment and to analyze the data obtained.
- (4) After experimenting, each student writes and submits a post-lab report and answers for the post-lab questions

individually.

- (5) After each experiment, clean and tidy up the experiment equipment and place used.
- (6) All descriptions, answers, explanations, and many others must be written in sentences and paragraphs. Any note-like description without a proper sentence and paragraph form will lead to point deduction.
- (7) The outline of the figures must be very clear. Any vagueness in color or contrast will lead to point deductions.

4.2 Academic Integrity Policy

- (1) The diversity of the class members ought to be respected. Any type of discrimination regarding ethnicity, nationality, language, personal orientation, etc., will not be tolerated.
- (2) (EC) All communications will be conducted in English as part of addressing the language diversity of all members. Other languages are not allowed unless permission is granted by the instructor.
- (3) Any type of misconduct (e.g., cheating, copying, possession of electronic devices during exam, refusing to follow instructions, dishonorable actions, other unauthorized activities not related to the course, etc.) will automatically result in an FAIL grade for the course. Furthermore, the student will receive official disciplinary punishment according to the Korea Science Academy Regulations.
- (4) Plagiarism and copying are strictly prohibited. Citations and references must be provided on a sentence-by-sentence basis whenever a source is used. Any sequence of six or more consecutive words taken from another source will be regarded as plagiarism. Note that paraphrasing is required whenever sources are referred to. All visual and technical materials (e.g., figures, equations, graphs, tables, and code) must be properly cited, and any reuse must comply with applicable permissions and licensing. Students must comply with all other regulations regarding academic integrity.
- (5) Experiments will be conducted in teams; however, all materials to be submitted (e.g., assignments, reports, questions, project files, and presentation files) must be prepared and submitted individually. Any misconduct - including plagiarism, copying, or other dishonest actions - will result in severe penalties, as described in [4.2 Academic Integrity Policy].
- (6) Using AI tools (e.g., ChatGPT) for academic support is permitted; however, students must verify and study the original sources cited or referenced. In addition, students must cross-check information using Google and multiple scholarly references. In this course, AI-generated content is not accepted as a citable source or reference. Submitting AI-generated content without proper citation to original sources will be regarded as plagiarism and unauthorized copying.
- (7) The use of AI tools (e.g., ChatGPT) to write or generate the contents of reports is strictly prohibited. The school may use plagiarism-detection tools (e.g., CopyKiller, Turnitin, GPTKiller, etc.). If a report exceeds a specified threshold (e.g., 15% similarity or higher), it will be considered plagiarism/unauthorized copying and noncompliance with the instructions.
- (8) Any late submission after the due date and time will not be accepted and will receive a score of 0 for the relevant criteria. If you use Google Classroom, make sure the assignment status shows "Submitted." Uploading a file without clicking "Submit" will not be considered a valid submission and will receive a score of 0 for the relevant criteria.

5. Evaluation

5.1 Evaluation Criteria

Pre-lab Assignment	Post-lab Report	Post-lab Questions	Project and Presentation	Attitude	Attendance	Sum
20	20	20	20	10	10	100

5.2 Evaluation Method

5.2.1 Pre-lab Assignment (20 points)

- (1) The pre-lab report must be prepared for every experiment and submitted before the assigned class begins.
- (2) The pre-lab report includes details on the introduction; the chemicals and instruments related to the experiment; a summary of the experimental procedures; the pre-lab questions; and other relevant items.
- (3) Pre-lab reports must be handwritten. The use of word-processing software (e.g., Microsoft Word, Pages, or Hancom Hangeul) is not permitted.
- (4) Figures and graphs must be drawn by hand or attached as printed materials.

5.2.2 Post-lab Report (20 points)

- (1) The post-lab report must be completed for every experiment and submitted before the next class begins.
- (2) The post-lab report includes quantitative results (e.g., calculation data, tables, graphs, etc.); qualitative results (e.g., observations, etc.); an overall summary of the experiment; an evaluation of the results; conclusions; references; and other relevant items.
- (3) The post-lab reports must be handwritten. The use of word-processing software (e.g., Microsoft Word, Pages, or Hancom Hangeul) is not permitted.
- (4) Figures and graphs must be drawn by hand or attached as printed materials.

5.2.3 Post-lab Questions (20 points)

- (1) The post-lab questions must be completed for every experiment and submitted before the next class begins.
- (2) The post-lab questions must be answered in detail.
- (3) The post-lab questions must be handwritten. The use of word-processing software (e.g., Microsoft Word, Pages, or Hancom Hangeul) is not permitted.
- (4) Figures and graphs must be drawn by hand or attached as printed materials.

5.2.4 Project and Presentation (20 points)

- (1) The project shall consist of a combination of computational (density functional theory, DFT) and literature studies on a topic that interests you and/or is applicable to your research. The results shall be presented to the class both orally (a 10–15 minute presentation during Week 9 of the dry lab) and in the form of a written report.
- (2) Students shall select a project topic of primary interest. For reference, potential topics include, but are not limited to, the following:
 - Molecular orbitals, bond energies, electron densities, electrostatic potentials, etc., of molecules
 - Structures and relative energies of different conformations of a molecule
 - Vibrational frequency and rotational constant predictions to explain spectra
 - Thermodynamic properties of a molecule (including heat of formation)
 - Charge density of a molecule and its implications for reactions governed by electrostatic effects
 - Energetics of a reaction (reaction enthalpy and activation barriers: energies of reactants, products, and transition states)
- (3) The presentations are scheduled for Week 9. Each student shall be allotted 10–15 minutes for the presentation.
- (4) The presentation must include sufficient background regarding the student's research project.
- (5) Following the project introduction, the presentation must specify the properties/systems simulated.
- (6) The methods used must be stated. Detailed explanations of the underlying theory are not required; however, the techniques shall be explicitly listed (e.g., geometry optimization at the B3LYP/6-31G(d,p) level).
- (7) The results and any conclusions drawn must be presented. Where feasible, results shall be compared with experimental work.
- (8) Limitations of the methods shall be addressed.

- (9) Students must be prepared to respond to questions from the instructor and other students.
- (10) The evaluation criteria are as follows: a project proposal (3 points) is due in Week 5, the presentation (7 points) will take place in Week 9, and the written report (10 points) is due in Week 10.

5.2.4 Attitude (10 points)

- (1) Assessment under the Attitude criterion will be subject to the instructor's discretion.
- (2) Students will start with 0 points in the Attitude criteria.
- (3) The attitude criterion will be evaluated based on the following: safe behavior, laboratory cleanliness, preparation for the lab (proper clothing, a lab notebook, and installation of LabJunior and Gaussian/GaussView), punctuality, respect for others, etc. Positive actions and behaviors may be rewarded with +0.5 points. Negative actions and behaviors will be penalized with -0.5 points per occurrence. Note that sleeping, disrupting the class, disobeying instructions, etc., may result in significant penalties.
- (4) A 1 point penalty will be applied for each violation of the tidiness regulations.

5.2.5 Attendance (10 points)

- (1) Students will receive 10 points for no unexcused absence and no late arrival.
- (2) A 0.5 point penalty will be applied for each late arrival to class.
- (3) A 1 point penalty will be applied for the first unexcused absence. The second unexcused absence will result in an additional 3 points penalty, and the third in a 5 points more penalty. (Total 9 of penalty for three unexcused absence).
- (4) Four or more unexcused absences will result in a FAIL grade for the course.

6. Criteria for Awarding Credits

A (100.00 ~ 85.00)			B (84.99 ~ 70.00)			C (69.99 ~ 55.00)			D (54.99 ~ 40.00)			F
A+	A0	A-	B+	B0	B-	C+	C0	C-	D+	D0	D-	
100.00	94.99	89.99	84.99	79.99	74.99	69.99	64.99	59.99	54.99	49.99	44.99	39.99
~	~	~	~	~	~	~	~	~	~	~	~	~
95.00	90.00	85.00	80.00	75.00	70.00	65.00	60.00	55.00	50.00	45.00	40.00	0

- (1) The course will follow an approximated absolute grading (approximated criterion-referenced grading) system. Grades will be assigned based on the over evaluation points that each student achieves.
- (2) Students with outstanding academic achievement may be recommended for the Academic Excellence Award and the Principal's Award, and their performance is recorded under the core competencies in the Subject Details. However, students with poor academic attitude or poor attendance will be excluded from consideration regardless of the achievement.

- Attendance for at least three-quarters of the total class hours is required to earn credit. An excessive number of excused or medical absences may result in a FAIL grade for the course, in accordance with the Korea Science Academy Regulations.
- For PASS/FAIL courses, if the instructor determines that a specific evaluation component is insufficient, an additional assessment may be conducted.
- The method for calculating the baseline score used to award credit to students with absences will be determined by the committee within the Department of Chemistry and Biology at the Korea Science Academy.
- If the baseline score cannot be calculated, the grading method will be determined by the committee within the Department of Chemistry and Biology at the Korea Science Academy.

7. Weekly Plans

Week (period)	Topic and Contents	Notes
Week 1 2.23.~3.1.	Orientation : Introduction, Lab Schedule, Method of Evaluation, and Safety Guideline of Chemistry Lab	Lab safety activity
Core Competencies	Subjective Comprehension, Scientific Thinking, Teamwork, Communication Skills	
Week 2 3.2.~3.8.	Laboratory Techniques	Peer Evaluation 3.2. Substitute Holiday
Core Competencies	Subjective Comprehension, Scientific Thinking, Communication Skills	
Week 3 3.9.~3.15.	Periodic Table and Periodic Law (Exp. 11)	
Core Competencies	Subjective Comprehension, Scientific Thinking, Teamwork, Ability to Explore, Creativity	
Week 4 3.16.~3.22.	Analysis of Atomic Line Spectra using Lab Junior (Exp. Dry Lab 3)	Laptop required for installing LabJunior
Core Competencies	Subjective Comprehension, Scientific Thinking, Teamwork, Ability to Explore, Creativity	
Week 5 3.23.~3.29.	Introduction to Computational Chemistry (Propose a topic for the project)	The project proposal deadline Laptop required for installing DFT Programs
Core Competencies	Subjective Comprehension, Scientific Thinking, Teamwork, Ability to Explore, Creativity	
Week 6 3.30.~4.5.	Computational Chemistry Project	Laptop required
Core Competencies	Subjective Comprehension, Scientific Thinking, Teamwork, Ability to Explore, Creativity	
Week 7 4.6.~4.12.	Computational Chemistry Project	Laptop required
Core Competencies	Subjective Comprehension, Scientific Thinking, Teamwork, Ability to Explore, Creativity	
Week 8 4.13.~4.19.	Midterm Exams (4.14.~4.17.)	4.13. Regular Classes

Week 9 4.20.~4.26.	Presentation for the Computational Chemistry Project	Laptop required
Core Competencies	Scientific Thinking, Teamwork, Ability to Explore, Creativity	Peer evaluation

Week 10 4.27.~5.3.	Transition Metal Complexes: Synthesis and UV-Vis Analysis (Exp. 36)	The project report deadline
Core Competencies	Subjective Comprehension, Scientific Thinking, Teamwork, Ability to Explore, Creativity	
Week 11 5.4.~5.10.	A Carbonate Analysis; Molar Volume of Carbon Dioxide (Exp. 13)	5.5. Children's Day
Core Competencies	Subjective Comprehension, Scientific Thinking, Teamwork, Ability to Explore, Creativity	
Week 12 5.11.~5.17.	Synthesis of Potassium Alum (Exp. 15)	
Core Competencies	Subjective Comprehension, Scientific Thinking, Teamwork, Ability to Explore, Creativity	
Week 13 5.18.~5.24.	Aspirin Synthesis and Analysis by ^1H -NMR Spectroscopy (Exp. 19)	
Core Competencies	Subjective Comprehension, Scientific Thinking, Teamwork, Ability to Explore, Creativity	
Week 14 5.25.~5.31.	Analysis of HCl (gas phase) by IR Spectroscopy	Meet at Changjo-gwan 5605 5.25. Substitute Holiday
Core Competencies	Subjective Comprehension, Scientific Thinking, Teamwork, Ability to Explore, Creativity	
Week 15 6.1.~6.7.	General Discussion: Class Improvements and Lab Cleaning	6.3. Local Election Day
Core Competencies	Scientific Thinking, Teamwork, Communication Skills	
Week 16 6.8.~6.14.	Make-up class week	-
Week 17 6.15.~6.19.	Final Exams (6.18.~6.19.)	-

8. Note

The course plan and policies may change at the instructor's consideration, depending on the given circumstances.